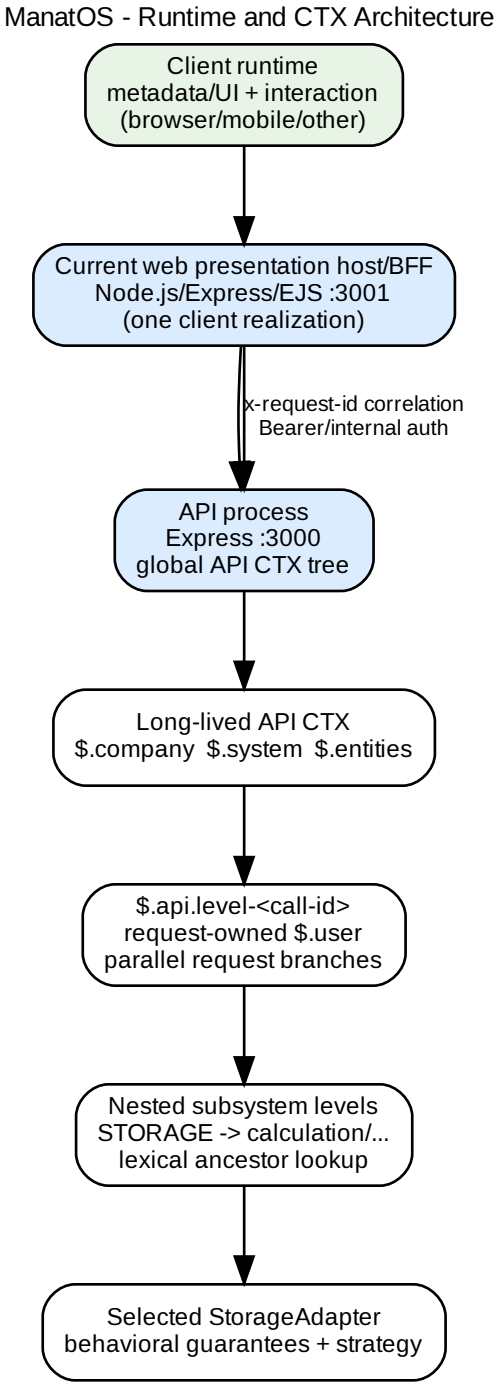
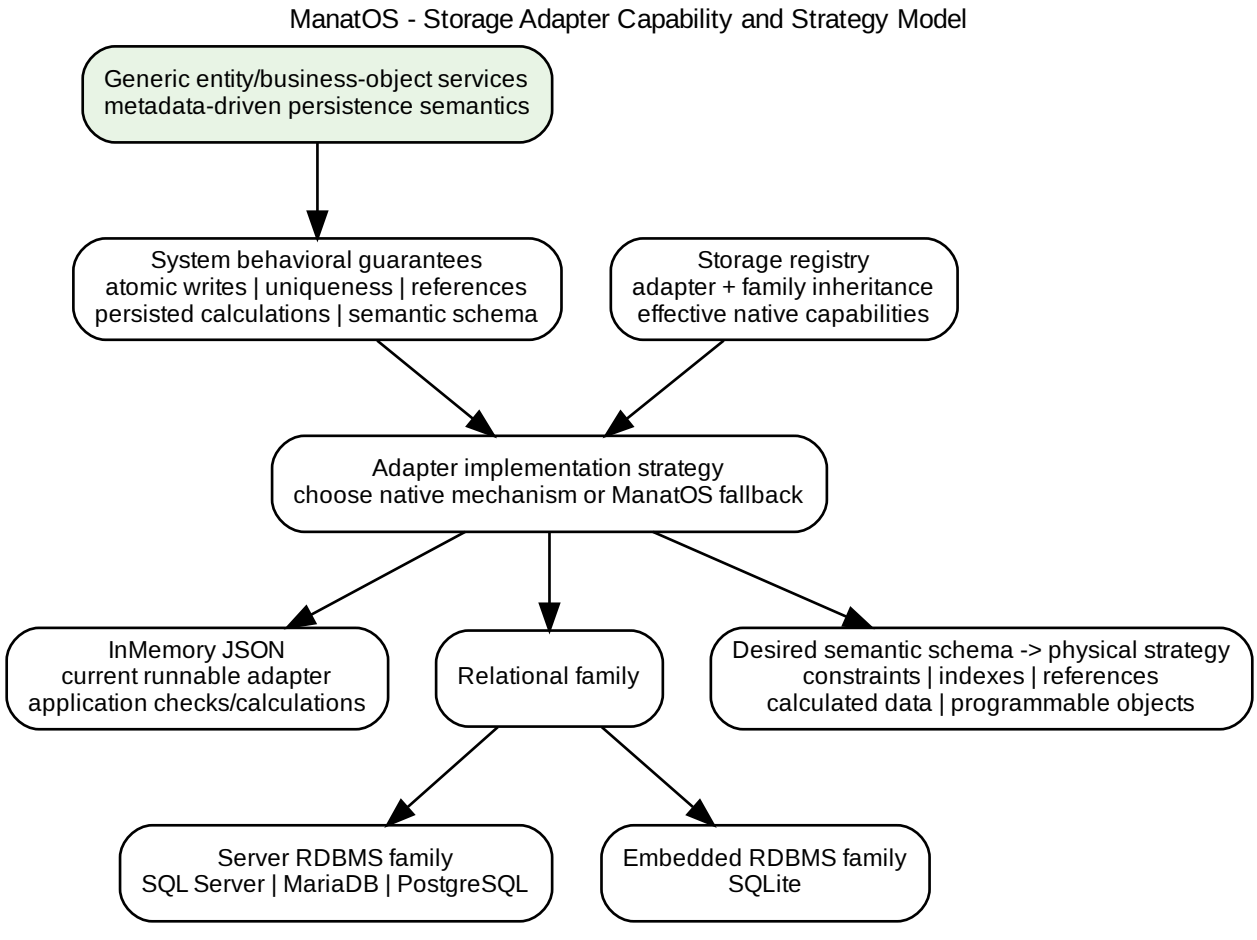
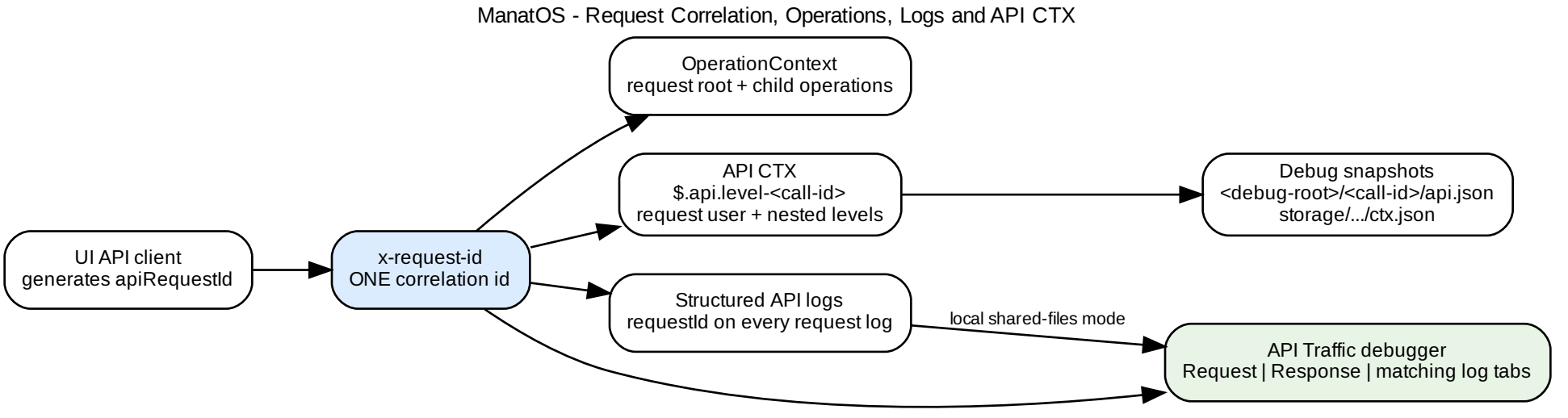
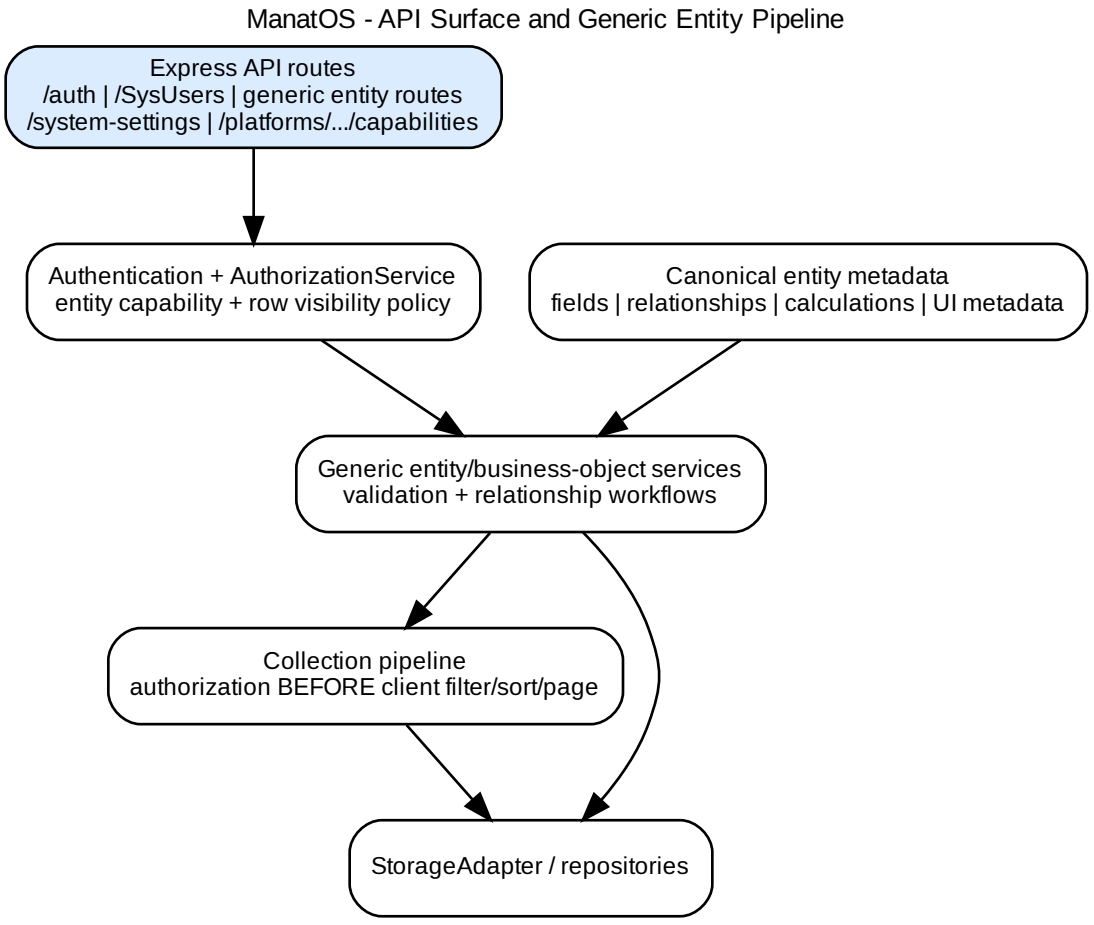
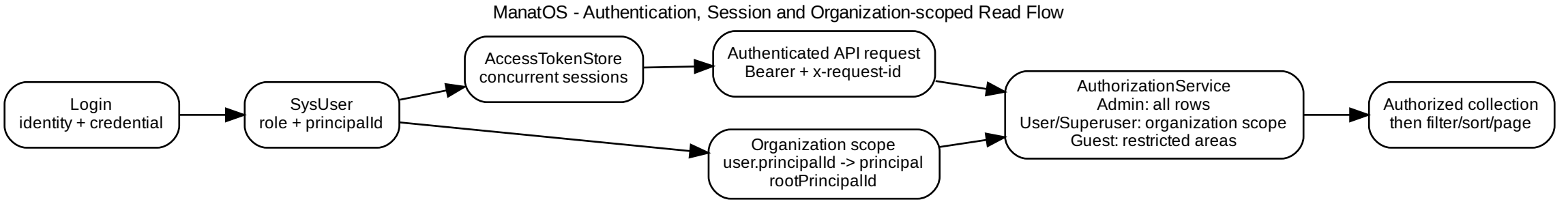

ManatOS - Company and protoCRM Domain Ownership
ManatOS / Company scope
Users + Account
Principals + organization hierarchy
Licenses
External Authentication Providers
SysUser
authenticates
optional principalId
identifies Person
parent/root
SysPrincipal
Person / Group / Company / System
parentId + rootPrincipalId
customer
SysLicense
principalId + platformId
optional applicationId
Platform
protoCRM
restriction
SysApplication
protoCRM-owned managed application
Future application-defined entities
reuse generic metadata/CTX/storage infrastructure
not automatically SysBO-prefixed

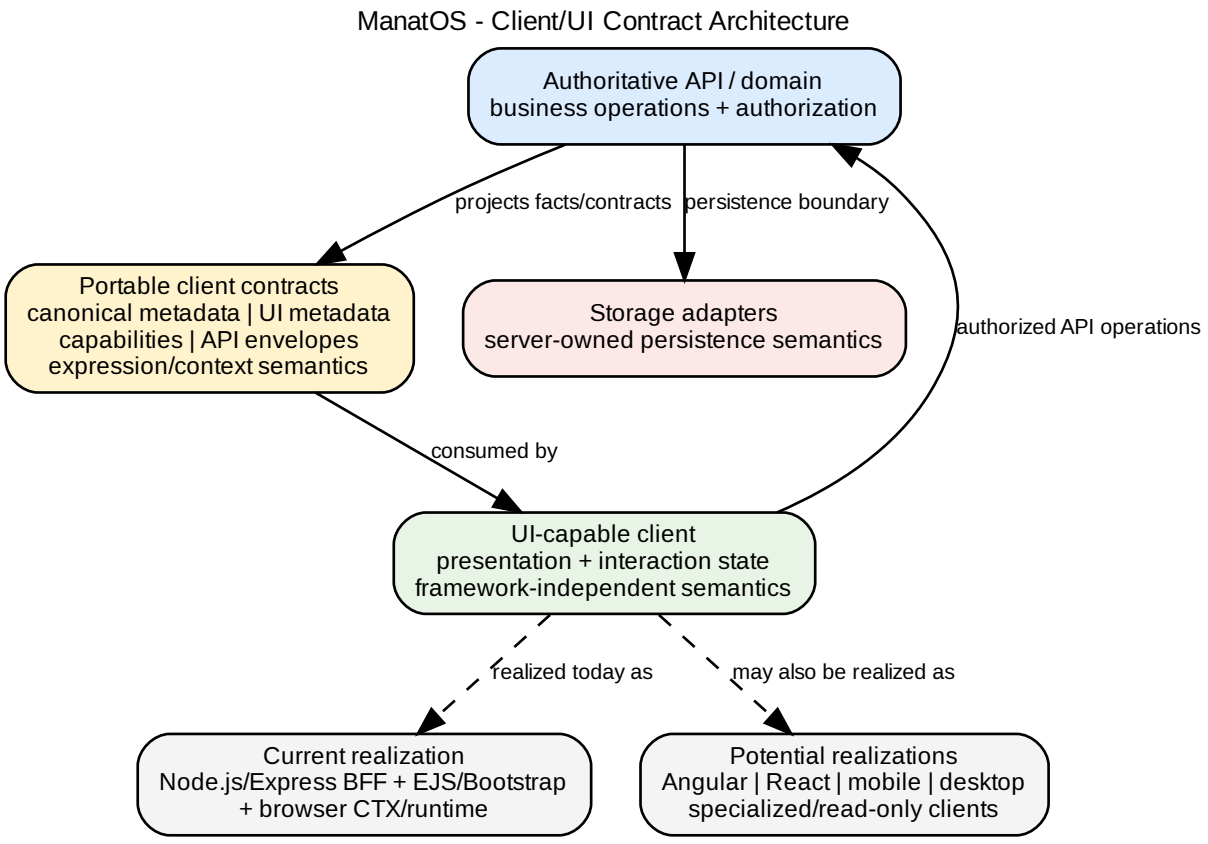

ManatOS - Runtime and CTX Architecture
Client runtime
metadata/UI + interaction
(browser/mobile/other)
Current web presentation host/BFF
Node.js/Express/EJS :3001
(one client realization)
x-request-id correlation
Bearer/internal auth
API process
Express :3000
global API CTX tree
Long-lived API CTX
$.company $.system $.entities
$.api.level-<call-id>
request-owned $.user
parallel request branches
Nested subsystem levels
STORAGE -> calculation/...
lexical ancestor lookup
Selected StorageAdapter
behavioral guarantees + strategy

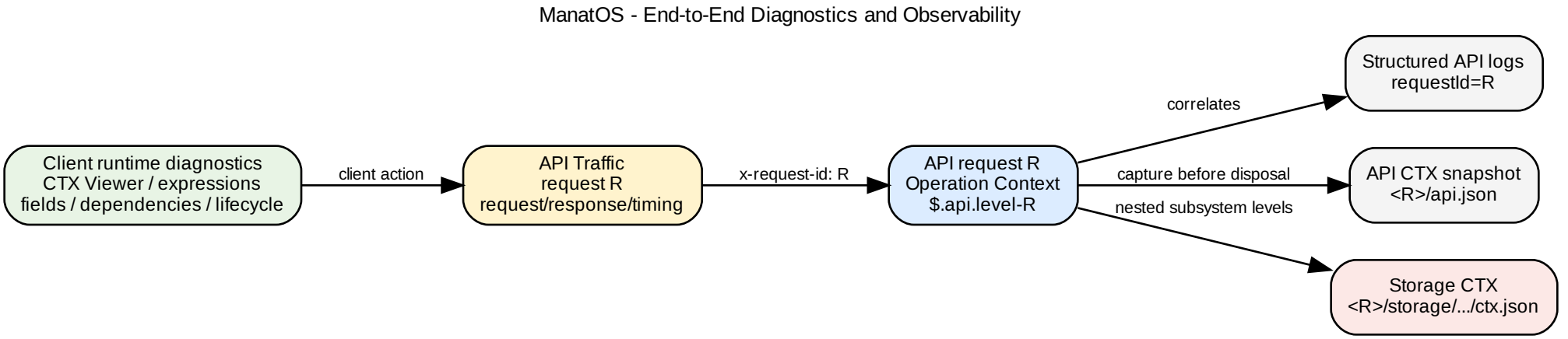

ManatOS - Storage Adapter Capability and Strategy Model
Generic entity/business-object services
metadata-driven persistence semantics
System behavioral guarantees
atomic writes | uniqueness | references
persisted calculations | semantic schema
Storage registry
adapter + family inheritance
effective native capabilities
Adapter implementation strategy
choose native mechanism or ManatOS fallback
InMemory JSON
current runnable adapter
application checks/calculations
Relational family
Desired semantic schema -> physical strategy
constraints | indexes | references
calculated data | programmable objects
Server RDBMS family
SQL Server | MariaDB | PostgreSQL
Embedded RDBMS family
SQLite


ManatOS - Request Correlation, Operations, Logs and API CTX
UI API client
generates apiRequestId
x-request-id
ONE correlation id
OperationContext
request root + child operations
API CTX
$.api.level-<call-id>
request user + nested levels
Structured API logs
requestId on every request log
Debug snapshots
<debug-root>/<call-id>/api.json
storage/.../ctx.json
local shared-files mode
API Traffic debugger
Request | Response | matching log tabs


ManatOS - API Surface and Generic Entity Pipeline
Express API routes
/auth | /SysUsers | generic entity routes
/system-settings | /platforms/.../capabilities
Authentication + AuthorizationService
entity capability + row visibility policy
Canonical entity metadata
fields | relationships | calculations | UI metadata
Generic entity/business-object services
validation + relationship workflows
Collection pipeline
authorization BEFORE client filter/sort/page
StorageAdapter / repositories


ManatOS - Authentication, Session and Organization-scoped Read Flow
Login
identity + credential
SysUser
role + principalId
AccessTokenStore
concurrent sessions
Authenticated API request
Bearer + x-request-id
Organization scope
user.principalId -> principal
rootPrincipalId
AuthorizationService
Admin: all rows
User/Superuser: organization scope
Guest: restricted areas
Authorized collection
then filter/sort/page


ManatOS - Client/UI Contract Architecture
Authoritative API / domain
business operations + authorization
projects facts/contracts
persistence boundary
Portable client contracts
canonical metadata | UI metadata
capabilities | API envelopes
expression/context semantics
Storage adapters
server-owned persistence semantics
authorized API operations
consumed by
UI-capable client
presentation + interaction state
framework-independent semantics
realized today as
may also be realized as
Current realization
Node.js/Express BFF + EJS/Bootstrap
+ browser CTX/runtime
Potential realizations
Angular | React | mobile | desktop
specialized/read-only clients


ManatOS - End-to-End Diagnostics and Observability
Client runtime diagnostics
CTX Viewer / expressions
fields / dependencies / lifecycle
client action
API Traffic
request R
request/response/timing
x-request-id: R
API request R
Operation Context
$.api.level-R
correlates
Structured API logs
requestId=R
capture before disposal
API CTX snapshot
<R>/api.json
nested subsystem levels
Storage CTX
<R>/storage/.../ctx.json